INCH-POUND
MIL-DTL-14852C (AR)
25 October 2007
SUPERSEDING
MIL-P-14852B (AR)
30 November 1988

DETAIL SPECIFICATION
PROJECTILE, 105MM, ILLUMINATING, M314A3
METAL PARTS ASSEMBLY

Inactive for new design after 9 July 1996

This specification is approved for use by the U.S. Army Armament Research Development and Engineering Center (ARDEC), and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Projectile, 105MM Illuminating, M314A3 Metal Parts Assembly.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in section 3 or 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to the Commander, U.S. Army ARDEC, ATTN: AMSRD-AAR-QES-E, Picatinny, NJ 07806-5000 or emailed to ardecstdzn@conus.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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2.2 Government Documents.

2.2.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

TT-C-490	-	Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings

DEPARTMENT OF DEFENCE STANDARDS

MIL-STD-1168	Lot numbering of Ammunition and Ammunition Data Card
Mil-STD-1916	DoD Preferred Methods for Acceptance of Product

(Unless otherwise indicated, copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings and Publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC) DRAWINGS

9280423	-	Magnetic Particle Standard for M314A3 Body
9280424	-	Magnetic Particle Standard for Base Plug, 105MM, M314A3
9280429	-	Rotating Band Seat Tester
10534900	-	Projectile, 105MM, Illuminating M314A3 Metal Parts Assembly

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Copies of these drawings may be requested online at Pica.drawing.request@conus.army.mil or from U.S. Army ARDEC, ATTN: AMSRD-AAR-AIS-TD, Picatinny, NJ 07806-5000.)

2.3 Non Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM E 8	Tension Testing of Metallic Materials
ASTM E 10	Brinell Hardness of Metallic Materials
ASTM B 117	Salt Spray (Fog) Testing
ASTM E 1444	Magnetic Particle Examination, Standard Practice For
ASTM E 103	Rapid Indentation of Hardness Testing of Metallic Materials
ASTM D 3951	Standard Practice for Commercial Packaging

(Copies of this document are available online at http://www.astm.org or from ASTM International, 100 Barr harbor Drive, West Conshohocken, PA 19428-2959.)

ASME INTERNATIONAL

ASME B 46.1	Surface texture

(Copies of this document may be requested online at http://www.asme.org or from ASME Codes and Standards, 3 Park Avenue, MS/20N2, New York, NY 10016).

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this specification takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article inspection. When specified (see 6.2) a sample shall be subjected to first article inspection in accordance with 4.2

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3.2 Components and assemblies.

3.2.1 Blank, rotating band. The components and assemblies shall be in accordance with all requirements specified on drawing 8594851.

3.2.2 Base plug (prior to painting). The components and assemblies shall be in accordance with all requirements specified on drawing 10534903.

3.2.3 Base plug (after painting). The components and assemblies shall be in accordance with all requirements specified on drawing 10534903.

3.2.4 Body (prior to banding). The components and assemblies shall be in accordance with all requirements specified on drawing 10534902.

3.2.5 Body assembly (prior to painting). The components and assemblies shall be in accordance with all requirements specified on drawing 10534901.

3.2.6 Metal parts assembly. The components and assemblies shall be in accordance with all requirements specified on drawing 10534900.

3.2.7 Projectile, 105MM, M314 metal parts assembly destructive lot acceptance. The components and assemblies shall be in accordance with all requirements specified on drawing 10534900.

3.3 Ammunition lot numbering. Ammunition lot numbers shall be assigned in accordance with MIL-STD-1168.

3.4 Mechanical properties

3.4.1 Components. Mechanical properties specified on the applicable drawing shall apply throughout each component.

3.4.2 Rotating band. After application and subsequent removal from the projectile body, the rotating band shall have the following mechanical properties:

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Tensile Strength	-	30,000 pounds per square inch (psi) minimum
Elongation	-	15 percent minimum in 1 inch

3.4.3 Base plug. Mechanical properties specified on drawing 10534903, shall apply throughout the base plug.

3.4.4 Body. Mechanical properties shall be as specified on drawing 10534902.

3.5 Rotating band seating

3.5.1 Rotating band cleanliness. Prior to banding, the band seat shall be clean and free of oil, dirt, grease, rust or other foreign matter. Shot or sandblasting is not permitted for cleaning the band seat after knurling.

3.5.2 Band blank – gilding metal. The rotating band blank shall be in the form of a solid ring and shall be annealed to comply with the applicable specification. Annealing prior to banding shall not be necessary in the case of hot banding in excess of 800 degrees F.

3.5.3 Application of band blank. The rotating band blank shall be pressed into the band seat after final heat treatment of the projectile body. The band blank shall be applied in such a manner as not to distort the projectile body.

3.5.4 Sidewall contact. The band blank must contact the sidewall of the band seat for a minimum of eight-tenths of the band seat depth, measured from the outside surface of the projectile body to the rotating band undercut surface.

3.5.5 Sidewall gap. There shall be no gap between the rotating band and the band seat sidewalls at the outside of the surface of the projectile body after the rotating band has been machined to final dimensions.

3.5.6 Rotating band clearance. The diametrical clearance between the rotating band and the band seat shall not exceed 0.006 inch, when the destructive method of testing is used; or 0.009 inch when the non-destructive method of testing is used.

3.6 Metal defects

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3.6.1 Body. The body, including the cavity, shall be free of cracks, splits, cold shuts, inclusions, pipe, porosity, bursts, folds, seams and other metal defects.

3.6.2 Body cavity. The body cavity shall be free from scale, fins, burrs, draw marks, laminations, imbedded foreign matter, pits and sharp edges.

3.6.3 Base plug. The base plug shall be free from cracks, splits, cold shuts, inclusions, pipe, porosity, bursts, folds, seams and other metal defects.

3.7 Ballistic test. The projectile assembly shall reveal no evidence of metal parts failure when ballistically tested at excess pressure (see 4.4.5.1). Metal parts failure shall be defined as:

3.7.1 Breakup. There shall be no loss or breakup of metal parts components in the gun bore, in flight or downrange.

3.7.2 Rotating band failure. There shall be no loss or fanning of the rotating band.

3.7.3 Shearing of rotating band. There shall be no shear of rotating band as evidenced by complete lack of engraving.

3.8 Workmanship. The requirements for workmanship are as shown on the applicable drawings, in the applicable specifications and the following:

3.8.1 Surface finish. The requirements for surface finish are as detailed on the applicable component drawing.

3.8.2 Protective coating. The protective coating shall be in accordance with the applicable drawing and the specification listed thereon.

3.8.3 Threads. Threads shall be full and undamaged for the entire minimum length or depth as specified on drawing 10534902.

3.8.4 Burr and sharp edges. No part shall have a burr or sharp edge which might interfere with the assembly or function of the round or which might be injurious to personnel handling the item.

3.8.5 Foreign matter. No part or assembly shall contain dirt, grease, chips, oil, rust, corrosion or other foreign matter.

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3.8.6 Painting. Painting shall comply with the requirements of the applicable drawing. All paint shall be dry to the touch before packing for shipment. The paint shall be considered dry when the soft tacky condition no longer exists and the paint feels firm when touched with a finger using moderate pressure.

3.8.7 Protection of rotating band. The machined rotating band shall be protected from being nicked, burred or otherwise damaged in subsequent handling prior to shipment.

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VERIFICATION

TABLE I. Requirement/verification cross reference matrix.

Methods of Verification N/A – Not Applicable 1 – Analysis 2 – Demonstration 3 – Examination 4 – Test							Classes of Verification A- First Article B- Acceptance		
Section 3 Requirements	Description	Verification Method					Verification Class		Section 4 Verification
		N/A	1	2	3	4	A	B	
3.1	First article inspection				X	X	X	X	4.2
3.2	Components & assemblies	X							
3.2.1	Blank, rotating band				X	X	X	X	4.3.2.1
3.2.2	Base plug (prior to painting)				X	X	X	X	4.3.2.2/4.4.3
3.2.3	Base plug (after painting)				X	X	X	X	4.3.2.3
3.2.4	Body (prior to banding)				X	X	X	X	4.3.2.4
3.2.5	Body assembly (prior to painting)				X	X	X	X	4.3.2.5
3.2.6	Metal parts assembly				X	X	X	X	4.3.2.6/4.4.3
3.2.7	Projectile destructive lot acceptance				X	X	X	X	4.3.2.7
3.3	Ammunition lot numbering				X		X	X	4.3.1
3.4	Mechanical properties	X							
3.4.1	Components					X	X	X	4.4.1
3.4.2	Rotating band					X	X	X	**4.4.1**
3.4.3	Base plug					X	X	X	4.4.1
3.4.4	Body					X	X	X	4.4.1
3.5	Rotating band seating	X							
3.5.1	Rotating band cleanliness				X		X	X	4.4.3
3.5.2	Band blank – gilding metal				X X		X	X	4.4.3
3.5.3	Application of band blank				X		X	X	4.4.3
3.5.4	Sidewall contact				X		X	X	4.4.3
3.5.5	Sidewall gap				X		X	X	4.4.3

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TABLE I. Requirement/verification cross reference matrix - Continued

Methods of Verification N/A – Not Applicable 1 – Analysis 2 – Demonstration 3 – Examination 4 – Test							Classes of Verification A First Article B Acceptance		
Section 3 Requirements	Description	Verification Method					Verification Class		Section 4 Verification
		N/A	1	2	3	4	A	B	
3.5.6	Rotating band clearance				X		X	X	4.4.3
3.6	Metal defects	X							
3.6.1	Body				X	X	X	X	4.4.5
3.6.2	Body cavity				X	X	X	X	4.4.5
3.6.3	Base plug				X	X	X	X	4.4.5
3.7	Ballistic test					X	X	X	4.4.6
3.7.1	Breakup					X	X	X	4.4.6
3.7.2	Rotating band failure					X	X	X	4.4.6
3.7.3	Shearing of rotating band					X	X	X	4.4.6
3.8	workmanship	X							
3.8.1	Surface finish				X		X	X	4.3.2.1-4.3.2.7
3.8.2	Protective coating				X		X	X	4.3.2.1-4.3.2.7
3.8.3	Threads				X		X	X	4.3.2.1-4.3.2.7
3.8.4	Burr and sharp edges				X		X	X	4.3.2.1-4.3.2.7
3.8.5	Foreign matter				X		X	X	4.3.2.1-4.3.2.7
3.8.6	Painting				X		X	X	4.3.2.1-4.3.2.7
3.8.7	Protection of rotating band				X		X	X	4.3.2.1-4.3.2.7

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4.1 Classification of verification. The verification requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection.

4.2.1 First article quantity. First article verification shall be performed on the quantity of items as indicated in table II.

4.2.2 Inspections to be performed. The first article verification shall be performed in accordance with Table II.

4.2.3 First article rejection. If any item fails to comply with requirements, the first article sample shall be rejected.

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Table II. First article inspection

CLASSIFICATION	EXAMINATION OR TEST	NUMBER OF SAMPLES	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Rotating band blank (Dwg. 8594851)			
	Examination for defects	5	3.2.1	4.3.2.1
	Mechanical properties	5	3.4.2	4.4.1.2.3
	Base plug (prior to painting Dwg. 10534903)			
	Examination for defects	4	3.2.2	4.3.2.2
	Yield strength & percent elongation	2	3.4.3	4.4.1.2
	Macro etch	2	3.2.2	4.4.2
	Metal defective	2	3.4.3	4.4.5
	Base plug (after painting Dwg. 10534903)			
	Examination for defects	3	3.2.3	4.3.2.3
	Salt spray test	2	3.2.2	4.4.4
	Body (prior to banding Dwg. 10534902)			
	Examination for defects	4	3.2.4	4.3.2.4
	Metal defective	4	3.6	4.4.5
	Yield strength & percent elongation	2	3.4.4	4.4.1.2

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Table II. First article inspection - Continued

CLASSIFICATION	EXAMINATION OR TEST	NUMBER OF SAMPLES	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Body assembly (prior to painting Dwg. 10534901)			
	Examination for defects	4	3.2.5	4.3.2.5
	Band seating	4	3.5	4.4.3
	Projectile metal parts (Dwg. 10534900)			
	Examination for defects	25	3.2.6	4.3.2.6
	Ballistic tests (Note 1)			
	a. Excess pressure	15	3.7	4.4.6
	Salt spray test	2	3.2.6	4.4.4

Note 1. Ballistic testing will be performed subsequent to successful completion of all dimensional and visual inspections outlined in Table II

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4.3 Conformance inspection.

4.3.1 Inspection lot formation. Lot formation shall be in accordance with lot formation requirements of MIL-STD-1916, Acceptance by Tables. In addition, inspection lots of projectile shall contain:

a. Base plugs from not more than one forging manufacturer.
b. Projectile bodies from not more than one forging manufacturer.

4.3.1.1 Heat treatment lots. Heat treatment lots shall contain body forgings from not more than one heat of steel that has been heat treated in the same heat treating equipment, at the same temperature in one unchanged process. The heat treatment lot thus constituted shall be the basis for the selection of samples for the mechanical properties test hereinafter prescribed.

4.3.1.2 Stress relieved lots. Stress relieved lots shall contain cold shaped bodies from not more than one heat of steel that have been cold shaped by the same process, utilizing the same tooling and equipment and are stress relieved at the same temperature for the same length of time. The stress relieved lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.3.1.3 Tubing lots. Bodies manufactured from tubing shall be fabricated from tubing lots containing not more than one heat of steel. The bodies shall be nosed and heat-treated using the same equipment and procedures. Lots of bodies thus constituted shall be the basis for the selection of samples for the mechanical properties test hereinafter prescribed.

4.3.1.4 Base plug lots. A base plug lot shall consist of not more than 5,000 base plugs or fraction thereof produced by one manufacturer in one unchanged process in accordance with the same drawings, same drawing revisions, same specification and same specification revisions. The lot thus constituted shall be the basis for selecting the samples for the mechanical properties test hereinafter prescribed.

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4.3.1.5 Miscellaneous lots. Small lots (such as bodies manufactured from abnormally small heats of steel or bodies that have become separated from previously delivered lots, etc.) may, at the discretion of the supplier, be grouped into miscellaneous heat treatment or stress relieved lots. Such lots shall be identified by a miscellaneous lot code number and processed in accordance with 4.3.1.1, 4.3.1.2, 4.3.1.3 or 4.3.1.4 as applicable. The miscellaneous lot thus constituted shall be the basis for the selection of samples for the mechanical properties test hereinafter prescribed.

4.3.1.6 Use. The heat treatment, stress relieved, base plug or miscellaneous lots constituted per 4.3.1.1, 4.3.1.2, 4.3.1.3, 4.3.1.4 or 4.3.1.5 shall be the bases for selecting samples for mechanical properties tests. Projectile bodies or base plugs in each heat treatment lot shall be identified to that particular lot until satisfactory completion of mechanical properties tests

4.3.1.7 Projectile Lot. Heat treatment, stress relieved, base plug or miscellaneous lots that have been determined to have complied with the yield strength and percent elongation requirements of the applicable drawing may be regrouped for assembly into projectile lots.

4.3.2 Classification of characteristics.

a. Sampling requirements. Inspection sampling requirements for critical, major and minor characteristics are defined in MIL-STD-1916. Unless otherwise specified, inspection level IV shall be used for all characteristics defined as majors and inspection level II for all minor characteristics. Critical characteristics shall be addresses in accordance with MIL-STD-1916.

b. Conformance inspection. Conformation inspection shall be performed accordance with paragraph 4.3.2.1 through 4.3.2.7. For all conformance inspections the same sample specimen may be used for all non-destructive examinations or tests.

c. Alternative inspections. Alternative conformance inspections shall be submitted and approved in accordance with MIL-STD-1916

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d. Automatic gaging. Facilities equipped with Government furnished automatic gaging capable of performing 100 percent inspection of any characteristic shall utilize capability in accordance with contract.

e. Tool control dimensions. Dimensions marked tool control shall be gaged after eight hours maximum production on each machine in operation and prior to any tooling change. Where destruction of components is necessary to inspect these dimensions, measurements of the tool may be substituted provided there is an established correlation between the tool dimensions and the component dimensions prior to the start of production. There shall be adequate surveillance to assure that drawing requirements are being met throughout the life of the tool.

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.1	TITLE Blank, rotating, band	SHEET 1 OF 1	DRAWING NUMBER 8594851 NEXT HIGHER ASSEMBLY 10534901

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Inside diameter	Level IV	3.2.1	AIE
102	Thickness	Level IV	3.2.1	AIE
103	Width	Level IV	3.2.1	AIE
Minor				
201	Evidence of poor workmanship	Level II	3.8	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.2	**TITLE** Base plug (prior to painting)	**SHEET** 1 **OF** 2	**DRAWING NUMBER** 10534903
			NEXT HIGHER ASSEMBLY 10534900

CLASSIFICATION	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**INSPECTION METHOD REFERENCE**
Critical				
1	Metal defective	100%	3.6.3	4.4.5
Major				
101	Runout of force fit surface with large outside diameter	Level IV	3.2.2	AIE
102	Perpendicularity of front face of flange with force fit surface	Level IV	3.2.2	AIE
103	Diameter of force fit surface	Level IV	3.2.2	AIE
104	Diameter of large outside surface	Level IV	3.2.2	AIE
105	Depth of undercut max	Level IV	3.2.2	AIE
106	length of force fit surface	level IV	3.2.2	AIE
107	length total	level Iv	3.2.2	AIE
108	Thickness through base, min	Level IV	3.2.2	AIE
109	Width of undercut	Level IV	3.2.2	AIE
110	Finish on force fit surface improper	Level IV	3.2.2	Visual
111	Yield strength and percent elongation *	Level IV	3.5.3	4.4.1.2.2
112	Protective coating (salt spray) *	Level IV	3.2.2	4.4.4
113	Macro etch *	Level IV	3.2.2	4.4.2

NOTES. * Sample size as specified in applicable test method paragraph

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.2	TITLE Base plug (prior to painting)	SHEET 2 OF 2	DRAWING NUMBER 10534903
			NEXT HIGHER ASSEMBLY 10534900

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Major				
201	Angle at front of force fit surface	Level II	3.2.2	AIE
202	Length from front face to shoulder	Level II	3.2.2	AIE
203	Radii or chamfers missing or incorrect	Level II	3.2.2	Visual
204	Evidence of poor workmanship	Level II	3.8	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.3	TITLE Base plug (after painting)	SHEET 1 OF 1	DRAWING NUMBER 10534903
			NEXT HIGHER ASSEMBLY 10534900

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Force fit surface damaged	Level IV	3.2.3	Visual
102	Salt spray test	4.4.3.3	3.2.3	4.4.4
Minor				
201	Paint on force fit surface	Level II	3.2.3	Visual
202	Protective coating inadequate or damaged	Level II	3.2.3	Visual
203	Wax emulsion missing from indicated surface	Level II	3.2.3	Visual
204	Evidence of poor workmanship	Level II	3.8	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.4	TITLE Body (prior to banding)	SHEET 1 OF 4	DRAWING NUMBER 10534902
			NEXT HIGHER ASSEMBLY 10534901

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Metal defective	100%	3.6.1	4.4.5.1
Major				
101	Diameter of rotating band seat	Level IV	3.2.4	AIE
102	Width of rotating band seat	Level IV	3.2.4	AIE
103	Ribs missing from band seat	Level IV	3.2.4	Visual
104	scoring missing from band seat	Level IV	3.2.4	Visual
105	Evidence of sand or shot blast on band seat after scoring	Level IV	3.2.4	Visual
106	Runout of pitch diameter of thread with bourrelets and front face	Level IV	3.2.4	AIE
107	Runout of large cavity with bourrelets	Level IV	3.2.4	AIE
108	Runout of body with bourrelets	Level IV	3.2.4	AIE
109	Runout of base plug cavity with bourrelets	Level IV	3.2.4	AIE

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.4	TITLE Body (prior to banding)	SHEET 2 OF 4	DRAWING NUMBER 10534902
			NEXT HIGHER ASSEMBLY 10534901

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Major				
110	Perpendicularity of face of base with bourrelets	Level IV	3.2.4	AIE
111	Diameter at front of large cavity, min.	Level IV	3.2.4	AIE
112	Diameter of small cavity, min. for min. length	Level IV	3.2.4	AIE
113	Diameter of large cavity	Level IV	3.2.4	AIE
114	Diameter of front bourrelet, min	Level IV	3.2.4	AIE
115	Diameter of body (between rotating band and bourrelet)	Level IV	3.2.4	AIE
116	Diameter of rear bourrelet, min	Level IV	3.2.4	AIE
117	Diameter of chamfer, max (at rear small cavity)	Level IV	3.2.4	AIE
118	Depth of small cavity	Level IV	3.2.4	AIE
119	Length of taper section at front of large cavity, max.	Level IV	3.2.4	AIE
120	Thickness of wall, min. at small cavity	Level IV	3.2.4	AIE

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.4	TITLE Body (prior to banding)	SHEET 3 OF 4	DRAWING NUMBER 10534902
			NEXT HIGHER ASSEMBLY 10534900

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Major				
121	Pitch diameter of thread, max.	Level IV	3.2.4	AIE
122	Pitch diameter of thread, min.	Level IV	3.2.4	AIE
123	Minor diameter of effective thread, max	Level IV	3.2.4	AIE
124	Length of effective thread, min	Level IV	3.2.4	AIE
125	Yield strength and percent elongation	Level IV	3.4.4	4.4.1.2.1
Minor				
201	Location of band seat	Level II	3.2.4	AIE
202	Foreign matter on band seat	Level II	3.2.4	Visual
203	Diameter of nose outside	Level II	3.2.4	AIE
204	Depth of large cavity	Level II	3.2.4	AIE
205	Length to rear of forward bourrelet	Level II	3.2.4	AIE
206	Length to ogive basic diameter	Level II	3.2.4	AIE
207	Length total	Level II	3.2.4	AIE

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.4	TITLE Body (prior to banding)	SHEET 4 OF 4	DRAWING NUMBER 9206749
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Minor				
208	Diameter of base plug cavity	Level II	3.2.4	AIE
209	Depth of base plug cavity	Level II	3.2.4	AIE
210	Radii, chamfers or notches missing or incomplete	Level II	3.2.4	AIE
211	Evidence of poor workmanship	Level II	3.8	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.5	TITLE Body assembly (prior to painting)	SHEET 1 OF 1	DRAWING NUMBER 10534901
			NEXT HIGHER ASSEMBLY 10534900

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Runout of rotating band seat with bourrelets	Level IV	3.2.5	AIE
102	Diameter of rotating band	Level IV	3.2.5	AIE
103	Length from rear face to start of rotating band	Level IV	3.2.5	AIE
104	Mechanical properties of rotating band	4.4.1.2.3	3.4.2	4.4.1.2.3
105	Rotating band and band seat clearance	4.4.2	3.5	4.4.3
106	Sidewall gap	Level IV	3.5.5	AIE
Minor				
201	Diameter of rotating band undercut	Level II	3.2.5	AIE
202	Length of rotating band flat	Level II	3.2.5	AIE
203	Length of rotating band total	Level II	3.2.5	AIE
204	Length of rotating band undercut	Level II	3.2.5	AIE
205	Evidence of poor workmanship	Level II	3.8	Visual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.6	TITLE Metal parts assembly	SHEET 1 OF 1	DRAWING NUMBER 10534900 NEXT HIGHER ASSEMBLY None

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Paint in thread	Level IV	3.2.6	Visual
102	Paint on walls of base plug cavity	Level IV	3.2.6	Visual
103	Rotating band damaged	Level IV	3.2.6	Visual
104	Coating missing or inadequate	Level II	3.2.6	Visual
105	Diameter of bourrelets, max	Level IV	3.2.6	AIE
106	Salt spray	4.4.3.2	3.2.6	4.4.4
Minor				
201	Protective coating inadequate or damaged	Level II	3.2.6	Visual
202	Marking missing, incorrect or illegible	level II	3.2.6	Visual
203	Evidence of poor workmanship	Level II	3.8	visual

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Conformance inspection by classification of characteristics

PARAGRAPH 4.3.2.7	**TITLE** Projectile, 105MM, M314 metal parts assembly destructive lot acceptance	**SHEET** 1 **OF** 1	**DRAWING NUMBER** 10534900
			NEXT HIGHER ASSEMBLY –

CLASSIFICATION	EXAMINATION OR TEST	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**INSPECTION METHOD REFERENCE**
Critical				
1	Ballistic test failure(excess pressure); projectile break-up or component separation in-bore or in flight	4.4.5	3.7.1	4.4.6
2	Ballistic test failure (excess pressure); rotating band failure	4.4.5	3.7.2	4.4.6
Major				
101	Ballistic test failure (excess pressure); shearing of rotating band	4.4.5	3.7.3	4.4.6
Minor	None defined			

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4.4 Methods of inspection

4.4.1 Mechanical properties

4.4.1.1 Hardness. After final heat treatment but prior to machining, the Brinell hardness test shall be performed on the outside surface at the band seat location. The test shall be performed using equipment and procedures as prescribed by ASTM E 10, except that a 5-second minimum dwell shall be used. A direct reading hardness tester may be substituted for the Brinell scope (micrometer microscope) type, provided that the same size indenter ball and the same pressure is used and that 2000 body forgings will have been tested by both units to establish correlation. Continued correlation shall be assured by checking the hardness value of one body forging each hour by measuring the diameter of the indent with a Brinell scope, for comparison with the dial reading.

4.4.1.1.1 Heat treatment lots. In order to determine uniformity of heat treatment, each body forging in the heat treatment lot (as defined by 4.3.1.1) shall be subjected to a hardness test with a Brinell hardness tester or by any other method of hardness testing approved by the procuring activity.

4.4.1.1.2 Stress relieved lots. To assess uniformity of process, twenty-five cold shaped bodies from the first batch of each stress relieved lot (as defined by 4.3.1.2) processed through the stress relieved furnace shall be subjected to a hardness test with a Brinell hardness tester or by any other method of hardness testing approved by the procuring activity.

4.4.1.1.3 Tubing lot. When mechanical properties of the tubing are altered by heat treatment or stress relief during manufacture of components, the uniformity of process shall be determined in accordance with 4.4.1.1.1 or 4.4. 1.1.2 as applicable.

4.4.1.1.4 Miscellaneous lots. Miscellaneous heat treatment lots shall be subjected to a hardness test in accordance with 4.4.1.1.1. Miscellaneous stress relieved lots shall be subjected to a hardness teat in accordance with 4.4.1.1.2.

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4.4.1.2 Yield strength and percent elongation

4.4.1.2.1 Body. From the samples provided by the heat treat or stress relieved lots two tension test specimens conforming to the largest obtainable standard “R” type specimen, as prescribed by ASTM E 8 shall be taken longitudinally from the opposite sides of the body at midwall, in such a manner that the center of the gage length of the machine test bar will fall in line coincident with the center of the rotating band seat. The number of specimens and their location are for the convenience of the contractor and Government and shall not restrict the contractor or Government from testing additional specimens at other locations to determine that the mechanical properties of the projectile body meet the drawing requirements. The criteria for performing this test shall be prescribed in ASTM E 8; except that the elongation shall be measured to the nearest thousandth (0.000) prior to conversion into percent elongation. Percent elongation shall be recorded to the nearest tenth (0.0) percent.

4.4.1.2.1.1 Heat treatment lots. Two projectile bodies, one of the hardest and one of the softest as determined by the hardness test, shall be subjected to this test. Failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be a cause for rejection of the lot. At the option of the supplier, a production control sample may be utilized to establish a hardness range. When a production control sample is used, the hardness range thus established shall be maintained throughout the heat treatment lot. The size of the sample shall be determined by the contractor. Projectile bodies which fall outside of the established range may be re-heated to fall within the range or new mechanical properties may be taken from newly selected projectile bodies which will represent the true extremes of hardness in the heat treatment lot.

4.4.1.2.1.2 Stress relieved lots. Two projectile bodies, one of the hardest and one of the softest as determined by the hardness test form the first twenty-five projectiles, shall be subjected to this test. Failure of any tension test specimen to comply with the mechanical properties requirement of the applicable drawing shall be a cause for rejection of the lot.

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4.4.1.2.1.3 Tubing lots. Two projectile bodies, one of the hardest and one of the softest as determined by the applicable hardness test, shall be subjected to this test in accordance with 4.4.1.2.1.1.

4.4.1.2.2 Base plug. From the samples provided, one tangentially transverse tension test specimen conforming to the largest obtainable standard “R” type specimen as prescribed by ASTM E 8 shall be taken form each base plug. The centerline of the specimen shall be between .50 and .75 inches from the centerline of the base plug. Testing of additional specimens at other locations to determine that the mechanical properties of the projectile body meet the drawing requirements is applicable. The criteria for performing this test shall be prescribed in ASTM E 8; except that the elongation shall be measured to the nearest thousandth (0.000) prior to conversion into percent elongation. Percent elongation shall be recorded to the nearest tenth (0.0) percent.

4.4.1.2.2.1 Base plug lots. Two base plugs from each lot of base plugs (as defined by 4.3.1.4) shall be selected at random and subject to this test. Failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be a cause for rejection of the lot.

4.4.1.2.2.2 Miscellaneous lots. Samples for tension test of miscellaneous heat treatment or stress relieved lots shall be selected in accordance with 4.4.1.2.1.1, 4.4.1.2.1.2 or 4.4.1.2.1.3 as applicable. Failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot. Tension tests shall be performed on each miscellaneous heat of steel before being grouped into miscellaneous lot.

4.4.1.2.3 Rotating band (gilding metal). The samples provided by 4.4.1.2.3.1 shall have the rotating band removed and carefully flattened without hammering so as to hold to a minimum any additional cold working. The specimen blank shall be machined so that all serration marks from the band seat are removed. A test specimen having a one inch gage length and a one-half inch gage width, and grip ends of sufficient length to accommodate the tensile machine shall be removed from the blank and tested in accordance with the procedures prescribed by ASTM E 8.

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4.4.1.2.3.1 Rotating band sample size. The sample rotating band for testing in accordance with 4.4.1.2.3 for compliance with 3.4.2 shall be provided as part of the sample taken from 4.4.3.3. The sample size, frequency and rejection criteria are the same as that described in 4.4.3.3.

4.4.2 Macro etch (base plug) one base plug from each standard heat treatment lot and one from each miscellaneous heat treatment lot shall be subjected to a Macro Etch test. The base plug shall be sectioned through its diameter and one of the sectioned surfaces prepared by grinding the surface to a flat. This part shall them be immersed in a macro etch solution of 50 percent hydrochloric acid and 50 percent water and heated to a temperature of 180 degrees F, minimum. The base part shall remain in the macro etch solution until flow lines become clearly visible. The part shall then be rinsed with water, dried and visually examined for compliance with Dwg. 10534903.

4.4.3 Band seating.

4.4.3.1 Clearance nondestructive method. The samples shall be tested for requirements in 3.5 utilizing the equipment and procedures prescribed Drawing 9280429. Five body assemblies from each 1-hour production per banding machine in operation shall be subjected to this test. Failure of one or more bands shall be cause for rejection of the assemblies represented by the failing sample.

4.4.3.2 Clearance destructive method. The body assembly selected for compliance with 3.5.6 need not be processed through the band machining line, although the band should have a light surface cut (0.020” max.) to provide a uniform surface for measurement. The diameter of each band shall be measured at three locations, sixty degrees apart, on the points along the cylindrical portion of their length, near the front and rear edges. This results in six (three pairs of two) diametrical measurements per band. The points of measurement shall be marked on the band and the projectile body adjacent thereto in such a manner as to permit proper body and band orientation after removal. The band shall be carefully removed so as to keep deformation to a minimum. After removal, the thickness of the band shall be measured at each of the marked points (a total of twelve measurements) and the diameter of the band seat corresponding to these points shall also be measured. Add

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to each of the diameters of the band seat the sum of the two corresponding two thicknesses of the band; the results obtained shall then be subtracted from the previously determined diameters over the band. One body assembly from each 8-hr production or fraction thereof per banding machine in operation shall be subjected to this test to determine compliance with 3.5.6. Failure to comply shall be cause for rejection of the assemblies represented by the failing sample.

4.4.3.3 Sidewall contact. The samples shall be sectioned by cutting through two planes normal to the longitudinal axis one-half inch forward and rear of the rotating band. The cylindrical portion containing the rotating band shall be sectioned approximately in half through a plane parallel to its longitudinal axis. One of the two halves shall be used for establishment of mechanical properties of the band; the other section shall be sectioned into two other parts, thus exposing four surfaces each of which reveals a view of the band swaged into the band seat. The exposed surfaces shall be subjected to a light grinding operation to remove the metal smear effects of saw cuts prior to measurements for depth of sidewall contact to determine compliance with the requirements of 3.5.4. One body assembly shall be selected at the beginning of each shift (not to exceed ten hours) from each banding machine to determine compliance with 3.5.4 and also 3.4.2 (see 4.4.1.2.3.1). If, for any reason there is a change in operator, machinery or material (by lot number) during a shift, another sample shall be taken. Failure to comply with the requirements of the specification shall be cause for rejection of all material processed since the previous sample.

4.4.4 Protective coating. A salt spray test shall be performed on the projectile body and base plug utilizing the equipment, methods and procedures specified in ASTM B 117 and in accordance with the applicable drawings and specifications. Failure to comply shall be cause for rejection of the production represented by the sample.

4.4.4.1 Cleaning and application of protective coating. The sample size and acceptance criteria for inspection of the cleaning of the item and the application of the protective coating shall be as specified in the applicable specification. Failure to comply with the requirements shall be cause for taking the corrective action prescribed therein.

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4.4.4.2 Projectile body. Salt spray testing shall be conducted on production projectile bodies after protective coatings have been applied per the requirements of Dwg. 10534900. Sampling shall be in accordance with TT-C-490. When on reduced sampling (twice per week) one sample shall be taken during the first 30 minutes of production and the second taken during the last day of production that week.

4.4.4.3 Base plug. Salt spray testing shall be conducted on production base plugs after protective coatings have been applied per the requirements of Dwg. 10534903. Sampling shall be in accordance with TT-C-490. When on reduced sampling (twice per week) one sample shall be taken during the first 30 minutes of production and the second taken during the last day of production that week.

4.4.5 Magnetic particle inspection.

4.4.5.1 Projectile body. Heat treated or stress relieved projectile bodies after final machining but prior to banding shall be inspected by the magnetic particle inspection technique. The procedures for performing this test shall be as specified in ASTM E 1444 for wet fluorescent method with indirect magnetization by continuous method or the residual method. The entire body inside and outside shall be inspected after inducing a circular field. The magnetic particle standard per Dwg. 9280423 shall be utilized to establish inspection sensitivity. Any body which fails to comply with the requirements of 3.6 shall be rejected. Projectile bodies which comply shall be stamped with the letter “M” as specified on the applicable drawing.

4.4.5.2 Base plug. Base plug after final machining shall be inspected by the magnetic particle technique. The procedures for performing this test shall be as specified in ASTM E 1444 for wet fluorescent method with indirect magnetization by continuous method or the residual method. The magnetic particle standard per Dwg. 9280424 shall be utilized to establish inspection sensitivity. Any base plug which fails to comply with the requirements of 3.6 shall be rejected. Base plugs which comply shall be stamped with the letter “M” as specified on the applicable drawing.

4.4.5.3 Suspension. The solid content of the suspension shall be maintained at the level recommended by the

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suspensoid manufacturer. The suspension shall be checked at the start of each day's production and at intervals of not over 4 hours of continuous operation, utilizing the procedures and equipment prescribed by section 4 of ASTM E 1444. When performing this test for solid content of the suspension, if it is determined that it does not comply, all bodies that have been tested and accepted since the last satisfactory test was performed shall be retested after the suspension has been brought to the proper level (see paragraph 6.3 and ASTM E 1444.)

4.4.6 Proving ground function tests.

4.4.6.1 Excess pressure. The metal parts assemblies inert loaded and assembled in a complete round, shall be fired in a gun for which the projectile is standard, the tube of which will have a minimum of 25 percent life remaining based on current gun tube condemnation criteria. A propellant charge adjusted to yield a mean pressure of 42,500 psi with no maximum individual exceeding 44,400 psi shall be used. Photographic equipment will be set up near the gun muzzle to record the flight of the projectile to determine compliance with 3.7. Recovery and examination of metal parts for any abnormal condition shall be accomplished as necessary.

4.4.6.1.1 Production sampling. Normal sampling shall be conducted until 3 consecutive lots have been accepted. Reduced sampling will then be instituted. Any failure occurring while on reduced sampling requires a return to normal sampling. Sample quantities for normal and reduced sampling on a per lot basis shall be as follows.

	Normal	Reduced
Phase: Excess pressure	5	2

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency, or within the Military Department's System Command. Packaging data

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retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The projectile metal parts contained in this specification are intended for use with Projectile, 105MM, Illuminating, M314A3.

6.2 Acquisition requirements.

a. Title, number and date of this specification.

b. Requirements for submission for first article sample.

c. Requirements for submittal of Acceptance Inspection Equipment (see 6.3).

d. Requirements for ammunition lot numbering.

e. Requirements for Certification of Non-destructive Test (NDT) personnel (see 6.6).

f. Submission of ballistic test data and ammunition data card (see 6.7).

6.3 Submission of contractor acceptance inspection equipment (AIE) designs for approval. Submit copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: AMSRD-AAR-QEM-A, Picatinny, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Visual examination qualification. When compliance with the applicable requirement is in doubt as a result of visual examination the characteristic may be measured or gaged to determine acceptability.

6.5 Visual standards. Visual standards for evaluation of metal defects will be established for each contract after

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the first month of production. However, establishment of visual standards will not waive the requirements for compliance with 3.6.

6.6 Certification of nondestructive test (NDT) personnel. All personnel operating or calibrating NDT equipment should be qualified and certified in accordance with NAS-410 “Certification and Qualification of Nondestructive Test Personnel”. Application for copies should be addressed to the Aerospace Industries Association of America, Inc, (AIA/NAS), 1000 Wilson Blvd, Suite 1700 Arlington, VA 22209. Qualification to other standard must be submitted to and approved by Commander, ARDEC, ATTN: AMSTA-AR-QAA-R, Picatinny Arsenal, NJ 07806-5000.

6.7 Submission of ballistic test data and ammunition data cards. In addition to the normal distribution of records for items procured by the Department of the Army, one copy of all ballistic test data and ammunition data cards should be forwarded to Commander, TACOM-ARDEC, ATTN: AMSRD-AAR-QEM-A, and AMSTA-AR-FSA-M, Picatinny Arsenal, NJ 07806-5000, and AMSMC-QAD-I Army Master Depository, Rock Island, IL 61299-6000.

6.8 Defect standards. Defect standards for metal defects will be established for each contract after the first month's production and must be approved by the technical engineering agency. However, the establishment of visual standards will not waive the requirement for compliance with 3.6.

6.9 Heat treatment. Heat treatment applied to the projectile body and base plug should be uniform throughout and should be conducted as to produce in each projectile body and base plug the mechanical properties specified by the applicable drawing. Metal forming involving plastic flow without subsequent heat treatment or stress relief is not permitted.

6.9.1 Hot forged – heat treatment method of manufacture. Components manufactured from forgings should be subjected to heat treatment to attain the mechanical properties specified by the applicable drawing. Heat treatment should consist of heating above the critical temperature, not to exceed 1600 degrees Fahrenheit, followed by an oil quench. After quenching, the component forging should be tempered without delay at a temperature above 750 degrees Fahrenheit

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which will produce the required mechanical properties. After removal from the tempering furnace, the component forging should be slowly cooled in air (forced air permitted) to a temperature on not more than 700 degrees Fahrenheit, after which cooling may be accelerated by water spray.

6.9.2 Tubing method of manufacturing. Body's manufactures from tubing should be heat treated or stress relieved after forming of the nose.

6.9.3 Base plug. The base plug should be heat treated by normalizing or oil quenching and tempering to obtain the specified mechanical properties in a transverse direction.

6.10 Protective finish. Details of the proposed procedure for zinc phosphate coating to be used by the contractor should be submitted in writing to the Government Technical Agency and written approval received prior to initiation of production. The proposed procedures should include materials, processes and equipment to be used by the contractor, detailed methods of control including limits for time, temperature, concentration and all other information pertinent to the processes. Approval of processes, materials, or equipment implies no guarantee of acceptance of the results obtained in use.

6.11 Process control of stress relief operation. A process control document covering the stress relief operation should be submitted to the technical agency through the contracting officer for review and approval prior to the initiation of production. This document should include descriptions of procedures, equipment, process parameters, monitoring practices and calibration procedures and frequencies to be used for the stress relief operation. Data should be included showing, based on temperature measurements, that the projectile bodies are at minimum temperature for the minimum time in accordance with 3.2.2 when using the process parameters outlined in the process control document. The process control document should be part of the contractor's quality program plan. Revisions should be submitted to the technical agency through the contracting officer for review and approval.

6.12 Critical characteristic controls. All production contracts should contain requirements for handling critical defects. As part of all proposals, suppliers should be

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asked to describe policies, procedures and controls for all operations associated with safety characteristics, how they are documented and maintained under the supplier's integrated management system

6.13 Process control dimensions. The contractor should prepare a process control plan for the dimensions marked process control. The plan should be submitted to the technical agency for approval prior to use. Dimensions to be gaged for process control are to be assured to a level IV.

6.14 Critical classification notes.

6.14.1 Ballistic lot acceptance test- projectile breakup or component separation. Metal parts separation in the gun or close to gun leads to leakage of energetic materials into the projectile and is likely to cause premature functioning of the round potentially resulting in injury or death for the Soldier within the Surface Danger Zone.

6.14.2 Rotating band failure. Loss of a rotating band in flight or in bore is likely to expose soldiers to hazards from fragments potentially resulting in injury or death for Soldiers in the field.

6.14.3 Metal defective. Metal parts defects, except rotating band gaps, are likely to lead to breakup of the projectile in the gun during firing. Fragments are likely to cause injury or death to the troop.

6.15 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes

6.16 Subject term (key word) listing

Ammunition
Artillery
Illuminating
Military Specification
Munition
Projectile, 105MM: Illuminating, M314A1, Metal Parts Assembly

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